SLF Crushers (T4)

Mechanisms

- Rotating System
- The reaper wall and bottom peg plate that converges flies to the center of the trap
- The flies drop into the death chamber via the rotating floor

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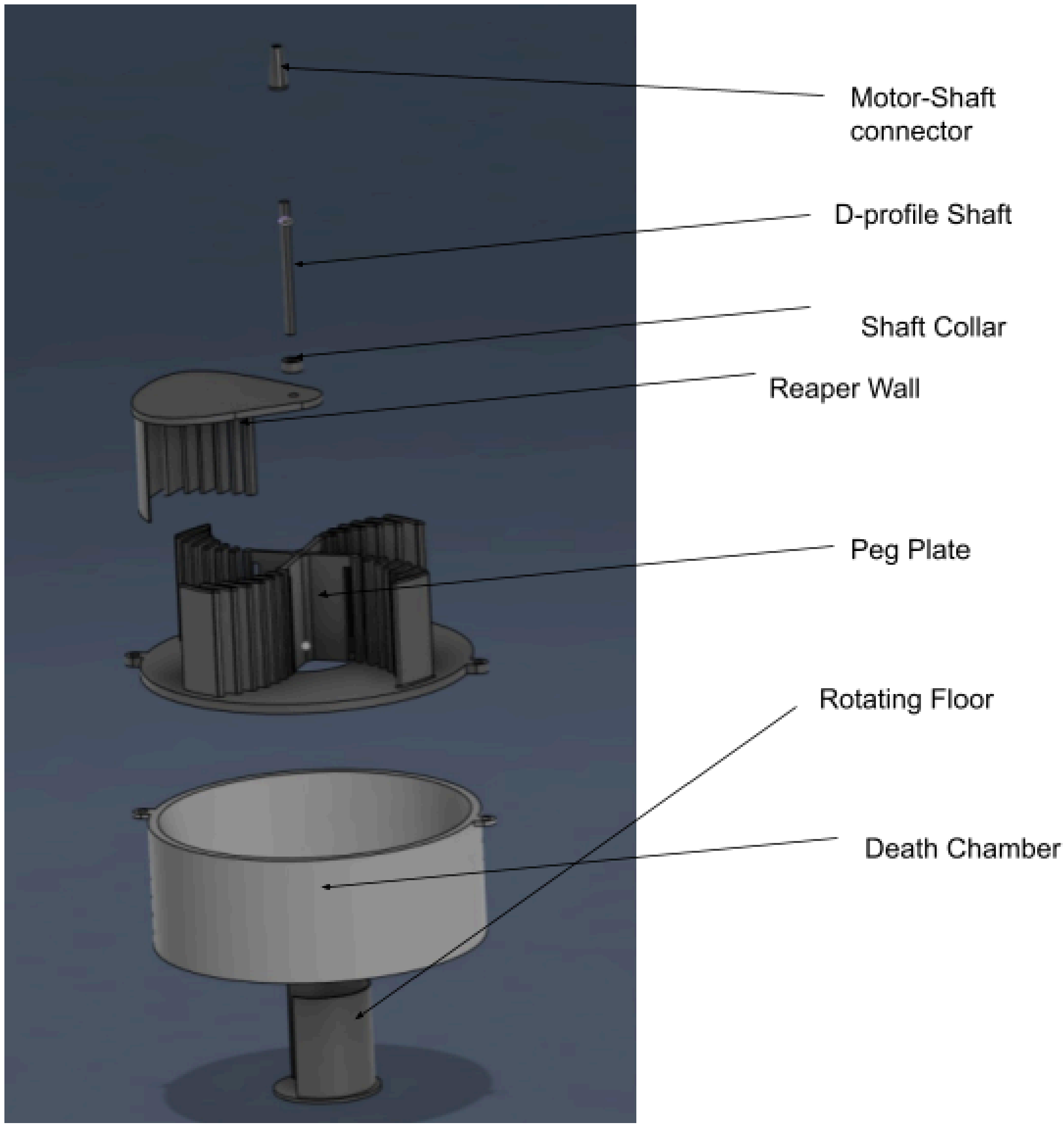
Design Documentation

Components List

Object Description	#	McMaster Code / Link to Amazon	Fabrication Details
D-Profile Rotary Shaft	1	8632T132	Length = 6in, Diameter = 0.25in Material: Carbon Steel Cut with Horizontal
Motor	1	Motor link	Voltage = 3V Speed = 5RPM Length = 27.5mm/1.08" D-Profile Shaft
Death Chamber (i.e. fly-collecting container)	1	N/A	3D Printed by RPL Material: PLA
Reaper Wall with 6 Pegs & External Wall (i.e., rotating paddle/blade)	1	N/A	3D Printed by RPL Material: PLA
Motor Attachment Lid	1	N/A	3D Printed by RPL Material: PLA
Peg Plate with 28 Pegs	1	N/A	3D Printed by RPL Material: PLA
Motor Shaft Connector	1	N/A	3D Printed by RPL Material: PLA
Rotating Floor	1	N/A	3D Printed by RPL Material: PLA



Explanation of How the Assembly Works

The peg plate pins into the housing attached to the death chamber. The pegs on the peg plate interlock with the pegs on the reaper wall, guiding the flies into the death chamber. The motor spins the D-profile shaft attached to the reaper wall, allowing the reaper wall to rotate. The motor is mounted onto the motor attachment lid using bolts. There is a motor shaft connector that will attach the motor's shaft to the D-profile shaft. There is a rotating floor attached to the D-profile shaft that rotates and creates an opening for the flies to fall into the death chamber. The sides of the rotating floors have walls that prevent the flies from going anywhere other than the opening that leads to the death chamber.

***Note -** Certain components including the motor-shaft connector and shaft collar, were not included in our functional prototype, because our tests did not require them at this stage.

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Design Test

Test	Part Being Tested	What It is Testing For	Procedure	Results	Conclusion/Next Iteration
Peg Movement with crumbled paper balls	Peg mechanism and fly channels	Whether the peg correctly guides and releases crumbled paper into the chamber without jamming or misalignment	Place crumbled paper balls into the entry point, manually spin the reaper wall, and observe whether balls drop cleanly into the death chamber	The crumbled paper balls were crushed by the pegs before they were guided to the center. We tested 5 times and got the same no matter where the ball was positioned on our peg plate.	We will make the peg path on the reaper wall have a greater curvature so that it guides the flies to the center rather than crushing them in the pegs.
Chamber Drop Test	Chamber opening and alignment	Whether crumbled paper balls reliably fall into the intended chamber without bouncing out or missing	Release paper balls from the standard position and observe landing behavior in the chamber	Yes, the crumbled papers fell into the death chamber reliably. The paper ball reliably dropped into the death chamber 5/5 times.	We plan on increasing the diameter of the hole that allows them to fall down because it currently only fits one fly (crumbled paper ball) at a time. Additionally, adding a roof over the center so that the flies wouldn't be able to fly out once trapped.
Manual Spin Test	Entire mechanical system (rotating components, connections)	Whether all moving parts function smoothly together under rotation, and if there is excessive friction or instability	Spin the system by hand at different speeds and observe movement, noise, and resistance	The system rotates, but it is slightly finicky. We did 10 revolutions, and every time it faced resistance– the reaper wall passing through pegs– it slowed or rotated the entire trap due to friction.	We are considering adding weights to the bottom so that the whole container doesn't move. Additionally, the pegs could use more tolerance so that they pass through each other more easily.

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Success Criteria

This project involves designing and testing a mechanical system that moves marbles using a peg mechanism into a chamber while rotating smoothly. The goal is to ensure the system operates reliably, efficiently, and is engaging during demonstration.

- **Reliable Paper Ball Transfer**
    - *What it assesses:* Whether paper balls consistently move through the peg mechanism into the chamber without jamming or missing.
    - *How it is measured:* Percentage of successful transfers over multiple trials (e.g., 9 out of 10 marbles successfully delivered).

- **Smooth Mechanical Operation**
    - *What it assesses:* How well the system rotates and whether parts move without excessive friction, resistance, or wobbling.
    - *How it is measured:* Observational rating (smooth vs. rough motion) and ability to complete full rotations without interruption.

- **Structural Stability**
    - *What it assesses:* Whether the system remains stable and intact during operation.
    - *How it is measured:* Observation of shaking, loosening parts, or failure during repeated use.

- **Efficiency of Design**
    - *What it assesses:* How quickly and effectively the system completes the marble transfer process.
    - *How it is measured:* Time taken for a marble to move from entry point to the chamber.

One criterion that is relevant to an exhibit-day demonstration is smooth mechanical motion. If we are demonstrating our device to people, it should operate smoothly with little friction, so that it is obvious that our device will be long-lasting and reliable.

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Image of Prototype: